



How to perform a cluster health check with a script in ONTAP

https://kb.netapp.com/on-prem/ontap/Ontap_OS/OS-KBs/How_to_perform_a_cluster_health_check_wit...

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Applies to

- ONTAP 9
- AFF systems
- FAS systems

Description

- Typically using [System Manager Dashboard](#), [Unified Manager health monitoring](#), [Active IQ Digital Advisor](#), and [Active IQ Config Advisor](#) provide the details necessary for actively monitoring overall ONTAP system health
- During system maintenance you might desire to perform manual health checks both before and after the maintenance

- If a more thorough review is necessary of your environment, engage [NetApp Services](#)

Procedure

Note: For ONTAP systems with AutoSupport enabled, comprehensive Health Check reports are available within [Active IQ Digital Advisor](#)

Main Commands

NOTICE

- These commands only perform checks and do not implement any changes
- Not all commands are available in all versions of ONTAP, refer to ONTAP documentation for the release in use for the cluster
- Commands that require a specific ONTAP release have a comment added to the end of the command line and will show in your SSH output but will not impact the execution of the command

1. Login to cluster via [SSH](#)
2. Prepare to save the SSH output as there may be many lines of scroll or consider the process for [recording a CLI session](#)
3. Enter advanced mode:

```
cluster1::> set -privilege advanced
```

4. Click the copy button (overlapping squares halfway down to the right of the contents below). Paste all of the clipboard into the SSH session (they can all be copied and pasted at once).

```
### START OF SCRIPT ###
set -rows 0 -prompt-timestamp inline
network interface check cluster-connectivity start -node * #Requires ONTAP 9.9.1+
```

```
### GENERAL ###
cluster kernel-service show -status-quorum out-of-quorum
cluster kernel-service show -status-avail !true
cluster kernel-service show -status-oper !operational
cluster peer health show -bypass-cache false -availability false
cluster ring show -online offline
cluster show -eligibility false
cluster show -health false
```

```

job show -state Failure|Error|Quit|Dead|Unknown
security key-manager key query -restored false #Requires ONTAP 9.6+
system chassis show -status !ok
system controller config show-errors -description !"sysconfig: There are no configuration errors."*
system controller environment show -status !ok
system controller memory dimm show -status !ok
system controller service-event show #Requires current supported ONTAP platforms
system fru-check show -fru-status !pass
system ha interconnect status show -is-link0-active !"-" -link0-status !up
system ha interconnect status show -is-link1-active !"-" -link1-status !up
system ha interconnect status show -link-status !"-" -link-status !up
system ha interconnect status show -ic-rdma !up
system health alert show
system health subsystem show -health !ok
system node environment sensors show -state !normal
system service-processor show -status !online
system services web node show -status !online

### STORAGE ###
storage aggregate show -state !online
storage aggregate show -raidstatus !*normal*
storage aggregate object-store show -object-store-availability unavailable #Requires ONTAP 9.2+
storage bridge show -status !ok
storage disk show -state !spare,!present -container-type !remote,!mediator
storage errors show
storage failover show -enabled false
storage failover show -partner-aggregates !false -state-description !"Non-HA mode"
storage failover show -possible false
storage port show -errors
storage shelf show -op-status !Normal

### NETWORK ###
network fcp adapter show -is-conn-established false -status-admin up
network interface show -is-home false
network interface show -status-oper down -status-extended !"This LIF is down because its Vserver is
stopped"*
network port ifgrp show -mode !singlemode -activeports !full
network port ifgrp show -mode singlemode -activeports !partial
network port reachability show -reachability-status !ok -expected-broadcast-domain !"-" #Requires
ONTAP 9.8+
network port show -health-status degraded
network port show -link !up -up-admin true

### DATA ###
lun show -state !online
snapmirror show -healthy false
volume show -in-nvfailed-state true
volume show -is-inconsistent true
volume show -state !online,!restricted -size !"-"
vserver cifs check -status down #Requires ONTAP 9.5+

```

```
vserver cifs show -status-admin down
vserver fcp show -status-admin down
vserver iscsi show -status-admin down
vserver nfs show -access !true
vserver show -type data -operational-state !running -subtype !dp-destination,!sync-destination
network interface check cluster-connectivity show -node * -loss !None -date-time >10m #Requires
ONTAP 9.9.1+
set -privilege admin -prompt-timestamp none

### END OF SCRIPT ###
```



5. Review the output of each command. Each command checks for an **unhealthy** condition.
 - a. If a command returns no results, then the unhealthy condition was not found
 - b. If results were returned, refer to the [Additional Information](#) section for troubleshooting tips to resolve the condition

Additional Checks

- To view client connections before and after maintenance see [How to View Active Client Connections by Protocol in ONTAP](#)
- Specific MetroCluster health check information can be found in the [MetroCluster Resolution Guide](#) under the Troubleshooting | MetroCluster Check decision tree
- More detailed information can be found in the [ONTAP Logs](#)
- Review [Performance monitoring and management overview](#) to confirm performance health
- Review [How to perform a security health check with a script in ONTAP](#)

Additional Information

Collection of Commands

This table includes a subset of commands to check system health, but should not be considered exhaustive

- **Note:** Items in **BOLD** require `set -privilege advanced` to be executed prior to running the command

- Not all AutoSupport sections are available in typical AutoSupport payloads and may require a [diagnostic AutoSupport](#) to be sent

General

Command	Expected Results	AutoSupport Section	Troubleshooting Tips
cluster date show	The date, time, and timezone are correct for all nodes in the cluster		Commands for managing the cluster time
cluster ha show	High Availability Configured: true (for 2 node clusters) High Availability Configured: false (for any other sized cluster)	CLUSTER-HA.XML	Enabling cluster HA and switchless-cluster in a two-node cluster
cluster kernel-service show -status-quorum out-of-quorum	There are no entries matching your query.	CLAM-NODE-STATUS.XML	Troubleshooting Workflow: Node is out of CLAM quorum in Clustered Data ONTAP 8.3 and ONTAP 9
cluster kernel-service show -status-avail !true	There are no entries matching your query.	CLAM-NODE-STATUS.XML	Troubleshooting Workflow: Node is out of CLAM quorum in Clustered Data ONTAP 8.3 and ONTAP 9
cluster kernel-service show -status-oper !operational	There are no entries matching your query.	CLAM-NODE-STATUS.XML	Troubleshooting Workflow: Node is out of CLAM quorum in Clustered Data ONTAP 8.3 and ONTAP 9
cluster peer health show -bypass-cache true -availability false	There are no entries matching your query.	CLUSTERPEER-HEALTH-MONITOR-	Troubleshooting Workflow: The cluster peer show

Command	Expected Results	AutoSupport Section	Troubleshooting Tips
network interface check cluster-connectivity show -node * -loss !None	There are no entries matching your query.	CACHE.XML	command reports an IP address as unhealthy Error: CLUSTER NETWORK DEGRADED
cluster ring show	The epoch, db-epoch, and db-trnxs match per unitname for all nodes.	CLUSTER-RING.XML	Verify cluster health
cluster show -eligibility false	There are no entries matching your query.	CLUSTER-INFO.XML	Verify cluster health
cluster show -health false	There are no entries matching your query.	CLUSTER-INFO.XML	Verify cluster health
event log show -time > 10m	A listing of the system events within the past 10 minutes.	EMS-LOG-FILE.GZ	FAQ: Overview of Event Management System for ONTAP 9
job show -state Failure Error Quit Dead Unknown	There are no entries matching your query.		Commands for managing jobs
security key-manager key query -restored false	There are no entries matching your query.	KEY-MANAGER-QUERY.XML	Managing NetApp encryption
system chassis show -status !ok	There are no entries matching your query.		system chassis show
system controller config show-errors -description !"sysconfig: There are no configuration errors."*	There are no entries matching your query.		system controller config show-errors
system controller environment show -status !ok	There are no entries matching your query.		system controller environment show
system controller fru led show -fru-state !off	There are no entries matching your query.		Procedure to troubleshoot fault LEDs on various ONTAP platforms

Command	Expected Results	AutoSupport Section	Troubleshooting Tips
	Note: This command is only supported for specific models and all other models will produce the following alert - "Command not supported on this platform". For clusters with mixed models it is recommended to run this command for each node individually with the -node parameter.		
system controller memory dimm show -status !ok	There are no entries matching your query.	MEMERR-VERBOSE.XML	How to troubleshoot correctable memory errors on FAS and AFF systems
	This table is currently empty. Note: This command is only supported for specific models and all other models will produce the following alert - "Command not supported on this platform. To view the attention LED states, use this command instead: system controller fru led show". For clusters with mixed models it is recommended to run this command for each node		
system controller service-event show			System fault LED is lit on FAS9000, FAS8200, FAS27xx, FAS26xx, AFF-A800, AFF-A700, AFF-A300, AFF-A220 or AFF-A200

Command	Expected Results	AutoSupport Section	Troubleshooting Tips
	individually with the -node parameter.		
system fru-check show -fru-status !pass	There are no entries matching your query.		system fru-check show
system ha interconnect status show -link-status !up	There are no entries matching your query.		High-availability pairs
system health subsystem show -health !ok	There are no entries matching your query.	SYSTEM-HEALTH-ALERT-PERSISTENT.XML	Responding to degraded system health
system node environment sensors show -state !normal	There are no entries matching your query.	PLATFORM-SENSORS.XML	Displaying environmental information
system service-processor show -status !online	There are no entries matching your query.	SP-INFO.XML	Managing a node remotely using the SP/BMC
Storage			
Command	Expected Results	AutoSupport Section	Troubleshooting Tips
storage aggregate show -state !online	There are no entries matching your query.	AGGR-STATUS-V, AGGR-STATUS-R	What is an aggregate?
storage aggregate show -raidstatus !*normal*	There are no entries matching your query.	AGGR-STATUS-V, AGGR-STATUS-R	What is an aggregate?
storage bridge show -status !ok	There are no entries matching your query.	STORAGE-BRIDGE	How to troubleshoot "storage bridge" errors?
storage disk show -state	There are no entries matching your query.	AGGR-STATUS-R, STORAGE-DISK	Removing a failed disk

Command	Expected Results	AutoSupport Section	Troubleshooting Tips
!spare,!present			
storage errors show	There are no entries matching your query.	STORAGE-FAULT	The storage errors show messages and their resolution
storage failover show -enabled false	There are no entries matching your query.	STORAGE_FAILOVER.XML	Commands for enabling and disabling storage failover
storage failover show -possible false	There are no entries matching your query.	STORAGE_FAILOVER.XML	Commands for enabling and disabling storage failover
storage port show -errors	There are no entries matching your query.	STORAGE-PORT.XML	ONTAP - How to re-enable a disabled SAS port
storage shelf show -op-status !Normal	There are no entries matching your query.	STORAGE-SHELF.XML	SAS Disk Shelves Service Guide
Network			
Command	Expected Results	AutoSupport Section	Troubleshooting Tips
network fcp adapter show -is-conn-established false -status-admin up	There are no entries matching your query.	FCP-ADAPTER.XML	FC Target adapter port link disconnected
network interface show -is-home false	There are no entries matching your query.	NETWORK-INTERFACE.XML	Logical Interface (LIF) is not on home port

Command	Expected Results	AutoSupport Section	Troubleshooting Tips
network interface show -status-oper down	There are no entries matching your query.	NETWORK- INTERFACE.XML	Note: LIFs located within a dp-destination or sync-destination SVM will show as operational down since the SVM is stopped. Logical Interfaces (LIF) are administratively up but operationally down (up/ down) after ONTAP node Reboot
network port ifgrp show -activeports !full	There are no entries matching your query.	IFGRPS.XML	Link aggregation and ifgrp troubleshooting
network port reachability show -reachability- status !ok	There are no entries matching your query.	REACHABILITY.XML	Repair port reachability
network port show -health-status degraded	There are no entries matching your query.	NETWORK- PORTS.XML	Network port in degraded state due to l2_reachability
network port show -link !up -up-admin true	There are no entries matching your query.	NETWORK- PORTS.XML	Monitoring the health of network ports
Data			
Command	Expected Results	AutoSupport Section	Troubleshooting Tips
lun show -state !online	There are no entries matching your	LUN.XML	Troubleshooting Workflow: LUN offline/out of space

Command	Expected Results	AutoSupport Section	Troubleshooting Tips
	query.		
snapmirror show -healthy false	There are no entries matching your query.	SNAPMIRROR.XML	SnapMirror basic network troubleshooting
volume show -in-nvfailed-state true	There are no entries matching your query.	VOLUME.XML	How does NVFAIL work with LUNs in clustered Data ONTAP 8?
volume show -is-inconsistent true	There are no entries matching your query.	VOLUME.XML	Inconsistent volume goes offline after a reboot/ takeover
volume show -state !online,!restricted	There are no entries matching your query.	VOLUME.XML	Commands for managing FlexVol volumes
vserver cifs check -status down	There are no entries matching your query.	CIFS-SERVER-BYNAME.XML	vserver cifs check
vserver cifs show -status-admin down	There are no entries matching your query.	CIFS-SERVER-BYNAME.XML	SMB/CIFS Reference
vserver fcp show -status-admin down	There are no entries matching your query.	FCP.XML	SAN Configuration Guide
vserver iscsi show -status-admin down	There are	ISCSI.XML	SAN

Command	Expected Results	AutoSupport Section	Troubleshooting Tips
	no entries matching your query.		Configuration Guide
vserver nfs show -access !true	There are no entries matching your query.	NFS-SERVERS-BYNAME.XML	NFS Configuration Power Guide
vserver show -type data -operational-state !running -subtype !dp-destination,!sync-destination	There are no entries matching your query.	VSERVER-INFO.XML	vserver show
Performance			
Command	Expected Results		Troubleshooting Tips
qos statistics volume latency show -iterations 10	Latency is not higher than acceptable for your environment shown in milliseconds (ms) or microseconds (us)		How to troubleshoot performance issues in ONTAP 9